



Linux Boot Camp

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Connecting

SSH

Windows users: PuTTY

(<http://www.chiark.greenend.org.uk/~sgtatham/putty/download.html>)

Mac/Linux users: Use 'ssh' command at terminal

ssh andrewid@shark.ics.cs.cmu.edu

Files

Windows: Tectia file transfer

Mac/Linux users: Use 'scp' command at terminal:

scp -r andrewid@unix.andrew.cmu.edu:~private/myfolder /some/local/folder

scp myfile.c andrewid@unix.andrew.cmu.edu:~private/myfolder

A message from Peter....

FOR THE LOVE OF ALL
THAT IS HOLY AND
SACRED, USE THE
SHARK MACHINES FOR
ALL OF YOUR
ASSIGNMENTS

Welcome!

```
$ ls  
$ cd private  
$ mkdir 15-213  
$ cd 15-213  
$ mv ~/Downloads/datalab-handout.tar .  
$ tar xvf datalab-handout.tar  
$ cd datalab-handout
```

Some Nice Terminal Shortcuts

- Pressing *tab* will **autocomplete** file and folder names!
- Control+C will **stop** execution of your current program!
- Control+R will let you **search** your command history!
- Control+L will **clear** your screen!
- `cmd arg1 ... argN > file1.txt` will put the output of `cmd` into `file1.txt`!
- `cmd arg1 ... argN < file2.txt` will pull the input of `cmd` from `file2.txt`!
- Use the **up** and **down** arrow keys to **scroll through your command history**!

Linux file pathing

- ~ is your **HOME DIRECTORY**
 - This is where you start from after you SSH in
 - On bash, you can also use \$HOME
- . is an alias for your **PRESENT WORKING DIRECTORY!**
- .. is the file path for the **PARENT DIRECTORY** of your present working directory!
- / is the file path for the **TOP-LEVEL DIRECTORY**
 - You probably won't use this too much in this class

ls <dir> - List

- Lists the files in the present working directory, or, if specified, `dir`.
- `pwd` tells you your Present Working Directory.

```
jbiggs@blueshark ~ $ ls
cover_letter.pdf  factorial.py  Movies      resume.pdf  test.wav
demo.py           foo2.py      Music       school      timer.py
Desktop           foo.txt      Pictures    solutions.py www
display.py        Fravic.pdf   private    src
Documents         Library      public     Templates
Downloads          Minecraft.jar  Public    test.py
jbiggs@blueshark ~ $ pwd
/afs/andrew.cmu.edu/usr10/jbiggs
jbiggs@blueshark ~ $
```

cd <directory> - Change Directory

- Changes your present working directory to directory
- Your main tool for navigating a unix file system

```
jbiggs@blueshark ~ $ ls
cover_letter.pdf  factorial.py  Movies      resume.pdf  test.wav
demo.py           foo2.py      Music       school      timer.py
Desktop           foo.txt      Pictures    solutions.py www
display.py        Fravic.pdf   private     src
Documents         Library      public      Templates
Downloads          Minecraft.jar  Public     test.py
jbiggs@blueshark ~ $ cd private/
jbiggs@blueshark ~/private $
```

mkdir <dirname> - MaKe DIRectory

- Makes a directory `dirname` in your present working directory.
- Directories and folders are the **same thing!**

```
jbiggs@blueshark ~ $ ls
cover_letter.pdf  factorial.py  Movies      resume.pdf  test.wav
demo.py          foo2.py      Music       school      timer.py
Desktop          foo.txt      Pictures    solutions.py www
display.py       Fravic.pdf   private     src
Documents        Library      public      Templates
Downloads        Minecraft.jar Public        test.py

jbiggs@blueshark ~ $ cd private/
jbiggs@blueshark ~/private $ mkdir 15-213
jbiggs@blueshark ~/private $ cd 15-213
jbiggs@blueshark ~/private/15-213 $
```

`mv <src> <dest>` - MoVe

- `cp` works in exactly the same way, but copies instead
 - for copying folders, use `cp -r`
- `dest` can be into an existing folder (preserves name), or a file/folder of a different name
- Also used to re-name files without moving them
- `src` can be either a file or a folder

```
jbiggs@blueshark ~ $ cd private/  
jbiggs@blueshark ~/private $ mkdir 15-213  
jbiggs@blueshark ~/private $ cd 15-213  
jbiggs@blueshark ~/private/15-213 $ mv ~/Downloads/datalab-handout.  
tar .
```

tar <options> <filename> - Tape ARchive

- Compression utility, similar to zip files on Windows
- For full list of options, see `man tar`
- As name suggests, was used on tapes!
- x - extract, v - verbose, f - file input
- All of our handouts will be in tar format.

```
jbiggs@blueshark ~/private/15-213 $ tar xvf datalab-handout.tar
datalab-handout/
datalab-handout/bits.c
datalab-handout/Makefile
datalab-handout/README
datalab-handout/btest.h
datalab-handout/btest.c
datalab-handout/bits.h
datalab-handout/decl.c
datalab-handout/tests.c
datalab-handout/fshow.c
```

chmod <permissions> <src>

- chmod is used to change the permissions of a file or directory.
- 777 will give all permissions
- src can be either a file or a folder

```
[sgoyal@makoshark datalab-handout]$ ls
bddcheck  btest    decl.c    Driverlib.pm  fshow.c  Makefile
bits.c    btest.c  dlc       driver.pl     ishow    README
bits.h    btest.h  Driverhdrs.pm  fshow       ishow.c  tests.c
[sgoyal@makoshark datalab-handout]$ chmod 777 btest
[sgoyal@makoshark datalab-handout]$
```

scp <src> <dest>

- Allows files to be copied to/from or between different hosts.
- The full path to the remote host needs to be specified
- Use the -r option to copy folders

```
[sgoyal@makoshark datalab-handout]$  
[sgoyal@makoshark datalab-handout]$  
[sgoyal@makoshark datalab-handout]$  
[sgoyal@makoshark datalab-handout]$ scp -r bovik@shark.ics.cs.cmu.edu:/afs/andrew  
.cmu.edu/usr/bovik/private/15213/datalab-handout some/local/folder
```

`rm <file1> <file2> ... <filen>` - ReMove

- Essentially the delete utility
- To remove an (empty) directory, use `rmdir`
 - To remove a folder and its contents, use `rm -rf`
 - **Please be careful, don't delete your project.**
 - **There is no "Trash" here. It's gone.**
 - **If someone asks you to use `rm -rf /` ignore them**

What's in a file? (using cat)

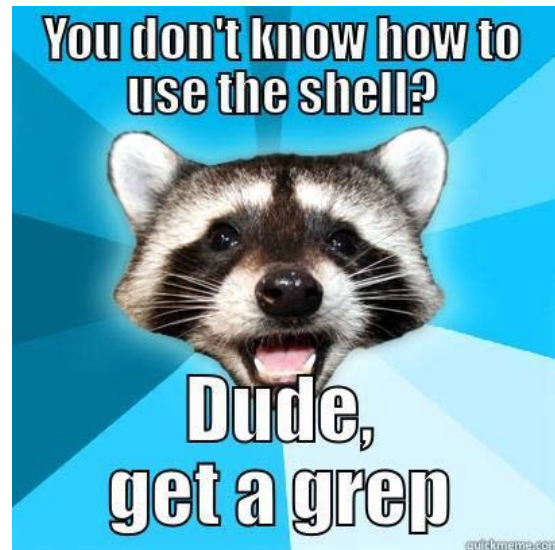
- `cat <file1> <file2> ... <filen>` lets you display the contents of a file in the terminal window.
 - Use `cat -n` to add line numbers!
- You can *combine* multiple files into one!
 - `cat <file1> ... <filen> > file.txt`
- Good for seeing what's in small files.
- Try `cat -n bits.c`. Too big, right?

What's in a file? (using less)

- `less <file>` will give you a scrollable interface for viewing large files **without** editing them.
 - To find something, use `/`
 - To view the next occurrence, press `n`
 - To view previous occurrence, press `N`
 - To quit, use `q`
- Try it: Type `"/isPower2"`

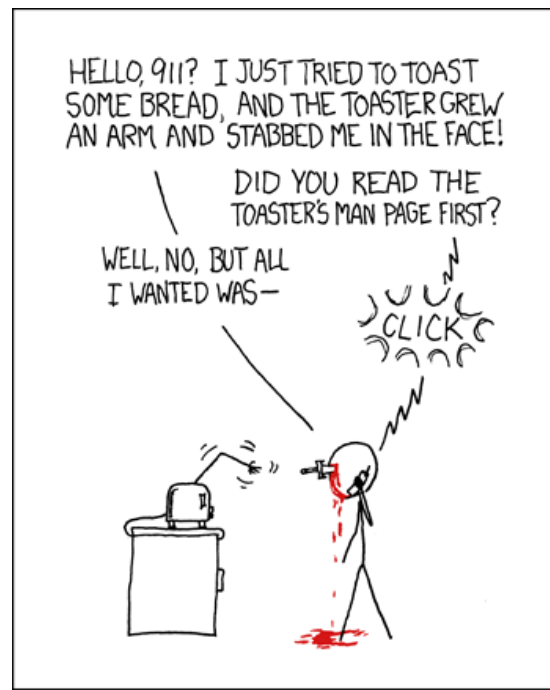
What's in a file? (using grep)

- `grep <pattern> <file>` will output any lines of `file` that have `pattern` as a substring
 - `grep -v` will output lines *without* `pattern` as substring
 - `grep -R` will search *recursively*
- Try it: `grep 'isPower2' bits.c`
 - `grep -v '*' bits.c`
 - `grep -R 'unsigned' .`

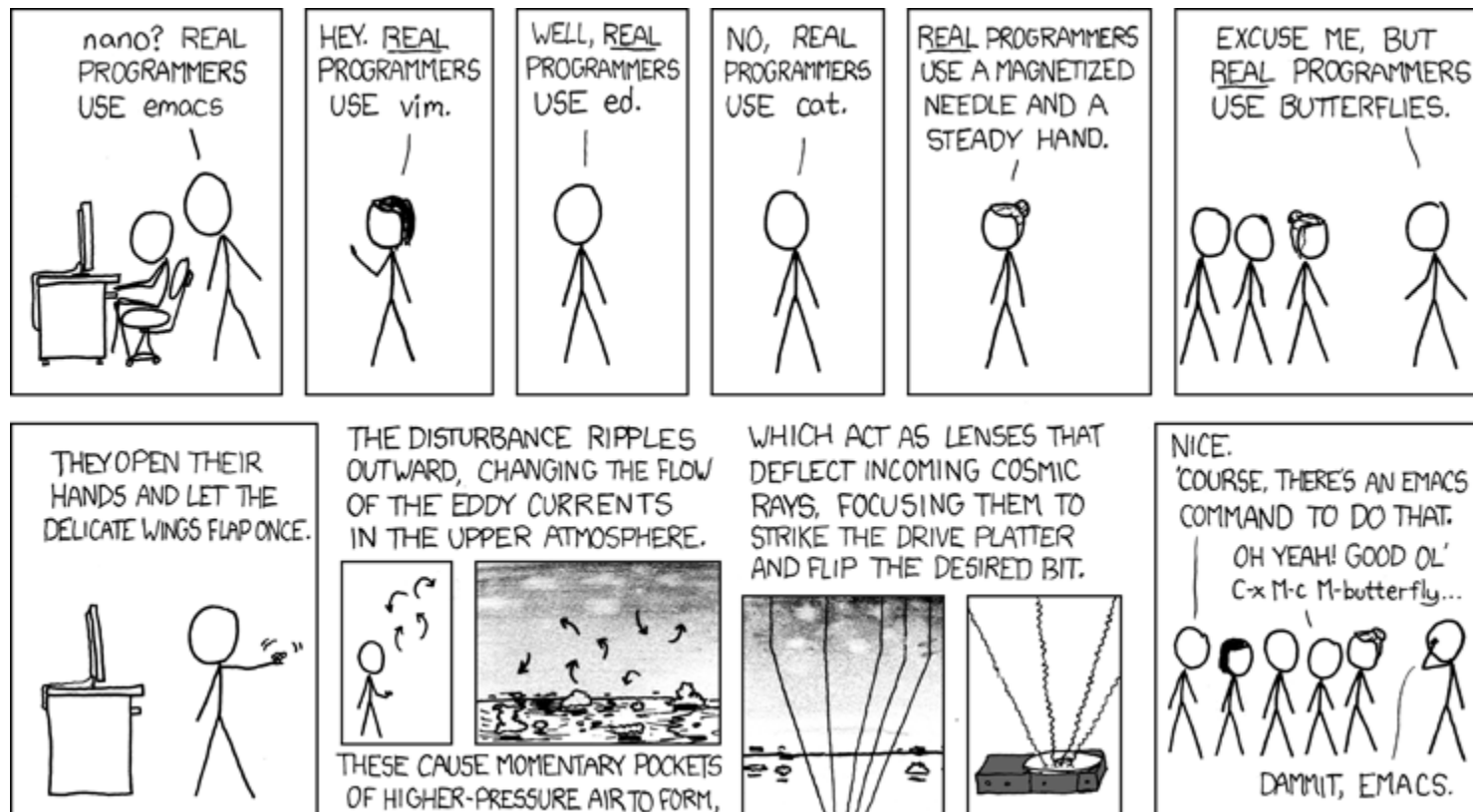


man <thing>

- What is that command? What is this C standard library function? What does this library do? Check to see if it has a man page!
- Pages viewed with less
- Try it!
 - man grep
 - man tar
 - man printf
 - man strlen



Editors (a touchy subject)



Vim (vi – improved) Basics

- Some different modes:
 - Normal mode:
 - The first mode you enter. Hit the **escape key** to return to this mode at any time
 - Everything entered here is interpreted as a *command*
 - Command-line mode:
 - Used for entering *editor commands* (necessary to save file & quit the editor)
 - Enter “:” in Normal mode to get to this mode
 - Insert mode:
 - To edit text
 - Enter “i” in Normal mode to get to this mode

Vim Basics

- Useful commands:
 - Copying/pasting/deleting lines:
 - yy (yank) or 5 yy (yank next 5 lines)
 - dd (delete) or 5 dd (delete next 5 lines)
 - p (paste)
 - Search (/search_string or ?search_string)
- Useful editor commands:
 - Write (w)
 - Quit (q) quit no-save (q!)

Vimrc File

- Stores vim configuration info
- Can make your editing experience even better!
- Notably:
 - Smart indentation
 - Line numbers
 - Changing tabs to default to 2 or 4 spaces
 - Colors
- To edit, type: `vim ~/.vimrc`

Vim colors

- Download a .vim color scheme file from the web (or make your own)
- Copy to ~/.vim/colors folder (make this folder if it doesn't exist)
- Some useful places to download color schemes:
 - <http://vimcolors.com/>
 - <http://cocopon.me/app/vim-color-gallery/>
- Makes your editor pretty!

```
1 require 'active_support'
2
3 module VimColors
4   class RubyExample
5     CONSTANT = /^[0-9]+ regex awesomes$/
6
7     attr_reader :colorscheme
8
9     # TODO: Bacon ipsum dolor sit amet
10    def initialize(attributes = {})
11      @colorscheme = attributes[:colorscheme]
12    end
13
14    def self.examples
15      # Bacon ipsum dolor sit amet
16      ['string', :symbol, true, false, nil, 99.9, 1..2].each do |value|
17        puts "it appears that #{value.inspect} is a #{value.class}"
18      end
19
20      {key1 => :value1, key2: 'value2'}.each do |key, value|
21        puts "the #{key.inspect} key has a value of #{value.inspect}"
22      end
23
24      %w[One Two Three].each { |number| puts number }
25    end
26
27    private
28
29    def heredoc_example
30      <<-SQL
31        SELECT *
32        FROM colorschemes
33        WHERE background = 'dark'
34      SQL
35    end
36  end
37 end
```

Jenna's Vimrc File

```
set tabstop=2  
set shiftwidth=2  
set expandtab
```

```
set viminfo='100,h  
colorscheme desertdocean  
set number  
syntax on  
filetype on  
filetype indent on  
filetype plugin on  
set smartindent
```

More resources on Vim

- A good intro tutorial:
<http://www.engadget.com/2012/07/10/vim-how-to/>
- An interactive tutorial: <http://www.openvim.com/>
- `man vim`
- Google

Commands related to 15-213

- `gdb`, the **GNU Debugger**, will be used for bomb lab.
- `objdump -d` displays the symbols in an executable.
- `gcc` is the **GNU C Compiler**.
- `make` reads a configuration file to run a series of commands. Often used for compiling your programs.
- We will provide other tools in the handouts as well

